

WASP-12b

R-1779

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_160223 · created 2026-08-02T23:21:34+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457442.71413 +/- 0.00025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.12 +/- 0.017
Transit depth (Rp/Rs)^2	1.44 +/- 0.4 [%]
Orbital Inclination (inc)	86.79 +/- 2.55
Airmass coefficient 1 (a1)	1.021 +/- 0.015
Airmass coefficient 2 (a2)	-0.0183 +/- 0.0069
Scatter in the residuals of the lightcurve fit is	1.47 %
Best Comparison Star	#1 - [525, 34]
Optimal Aperture	3.54
Optimal Annulus	9.04
Transit Duration (day)	0.1287 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.12 $\pm$ 0.017 vs 0.1178 (deviation 0.11 σ)
Duration fit vs published	0.1287 d vs 0.125 d (deviation 0.92 σ)
Mid-time O-C	1572.1 min
Depth SNR	6.2
Residual scatter	1.47 %

Light curve

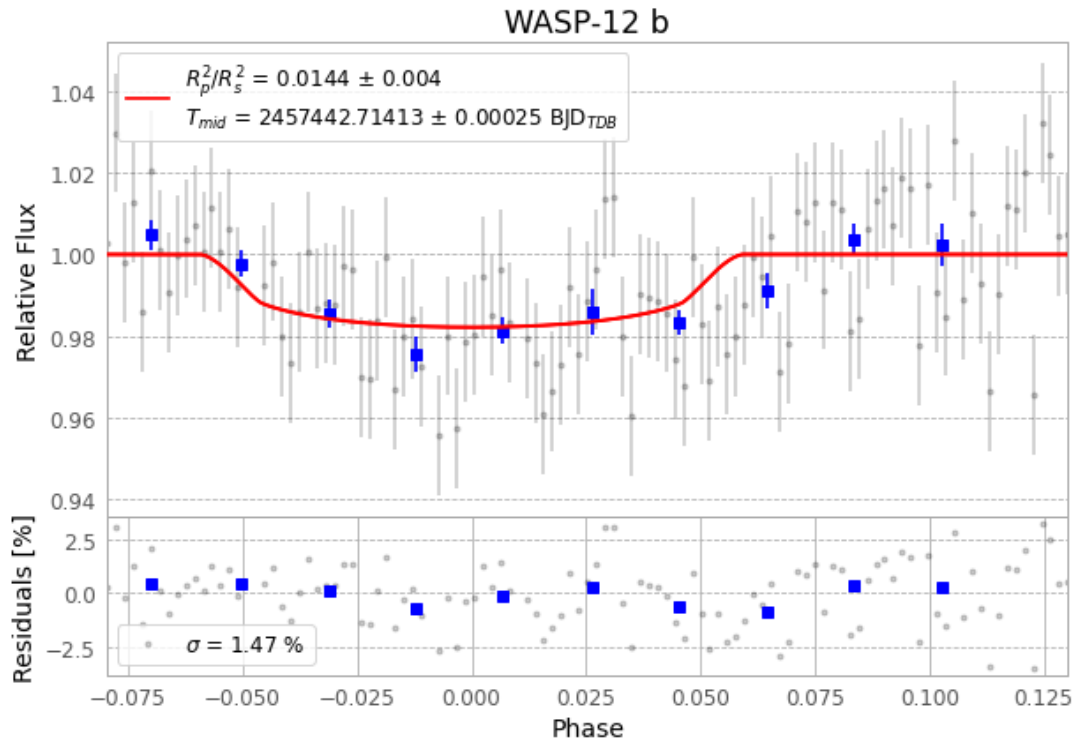


Figure 1 — FinalLightCurve_WASP-12 b_2016-02-23.png (SHA-256 281245d28b6a25c7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [339, 219], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2a499df4d1640a61...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

109 FITS frames (72 MB), WASP-12160224025712.FITS → WASP-12160224082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-160223.log (be53eea2164c...), FinalLightCurve_WASP-12 b_2016-02-23.png (281245d28b6a...), FinalLightCurve_WASP-12 b_2016-02-23.csv (ddc81a7dd97a...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 23:21Z.